

RAG: Naïve to Agentic

April 2026

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Knowledge Sharing #3

IPU, NECTEC

RAG

Full Credits to

- RAG deeplearning.io
- Ajarn. Natchanon Yon.
- freecodecamp
- Megawiz
- GiovanniPasq github
- Datacamp

PREREQUISITES

1. Python

- a. Data Type, Class , etc.
- b. LangChain, LangGraph, etc.

2. Basic LLMs knowledges

- a. Fine-tuning, Continue Pre-training (SFT, LoRA, DPO, etc.)
- b. Quantization

3. APIs

- a. GenAI - OpenAI (recommend), etc.
- b. Search - Serper (recommend), etc.

Course Roadmap

1. Why RAG ?
2. Basic RAG
3. RAG Improvement
4. Agentic RAG
5. Hands-On

!!! DISCLAIMER !!!

LLM IS MATH AND ART.

So, some methods can't be mathematically explained.

WHY RAG?

LLM Limitations

-  **Knowledge Cutoff**
 - Trained until a fixed date — knows nothing after
-  **No Real-Time Data**
 - Cannot access live prices, news, or current events
-  **Hallucination**
 - Confidently generates false information
-  **Private Data Blind**
 - Has never seen your company/private docs, PDFs, DB

Retrieval Augmented Generation (RAG)

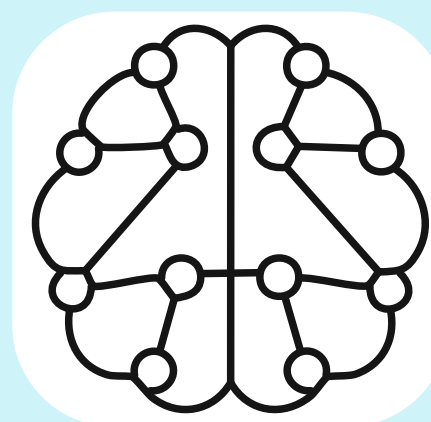
LLM →

Summarize Text

Generate Code

Rewrite Content

RAG further improves them



+



LLM

New
information

Example: Bangkok Traffic

? Question Asked to LLM:

- ***"Why is there so much traffic in Bangkok today?"***

Without RAG

"Bangkok is known for heavy traffic due to its large population, high car ownership rates, and rapid urban growth. The city's infrastructure has not kept pace..."

With RAG

"BLACKPINK announced a concert at Rajamangala Stadium tonight at 8 PM. An estimated 60,000 fans are expected, causing road closures on Ramkhamhaeng Road and surrounding areas..."

How ?

How ?

User Prompt

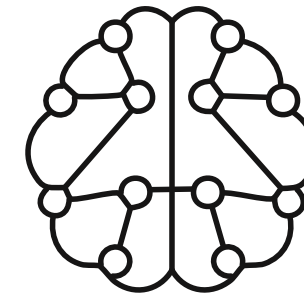
"Why is there so much traffic in Bangkok today?"



RAG system



Augmented Prompt



LLM



Response

BLACKPINK announced a concert at Rajamangala Stadium tonight at 8 PM. This weekend on April 24-26, 2024

Application of RAG

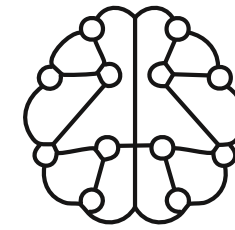
- **Company Chatbots**
- **Code Generation (LLM know your project)**
- **Specialized Knowledge (Legal and Medical)**
- **Personalized RAG**
- **etc.**

BASIC RAG

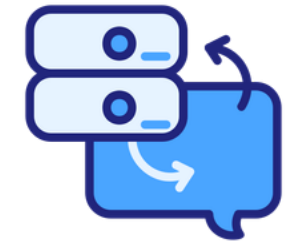
**Normal
LLM use**



Prompt



LLM

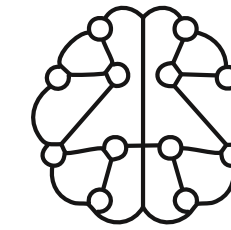


Response

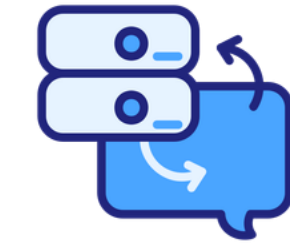
Normal
LLM use



Prompt

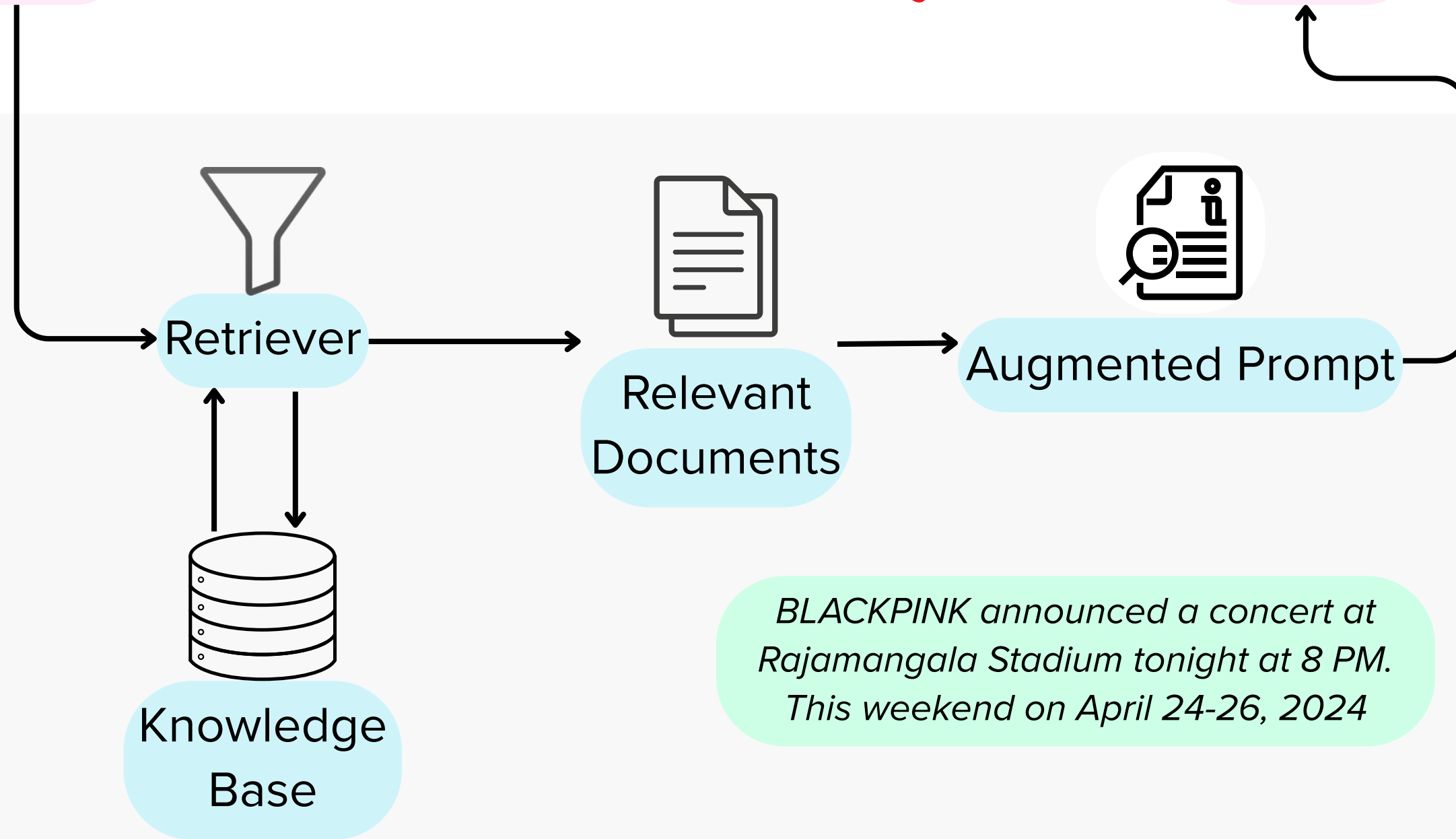


LLM



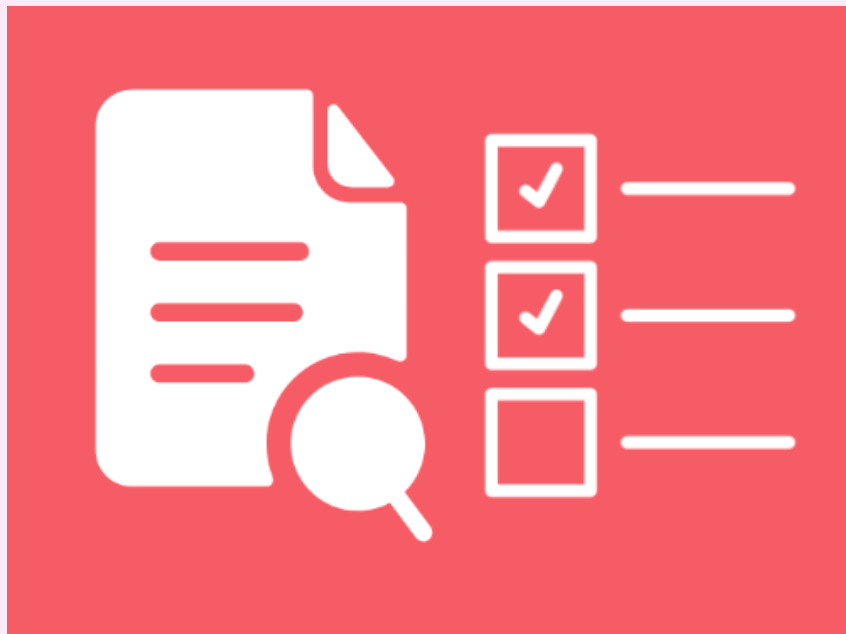
Response

RAG
System



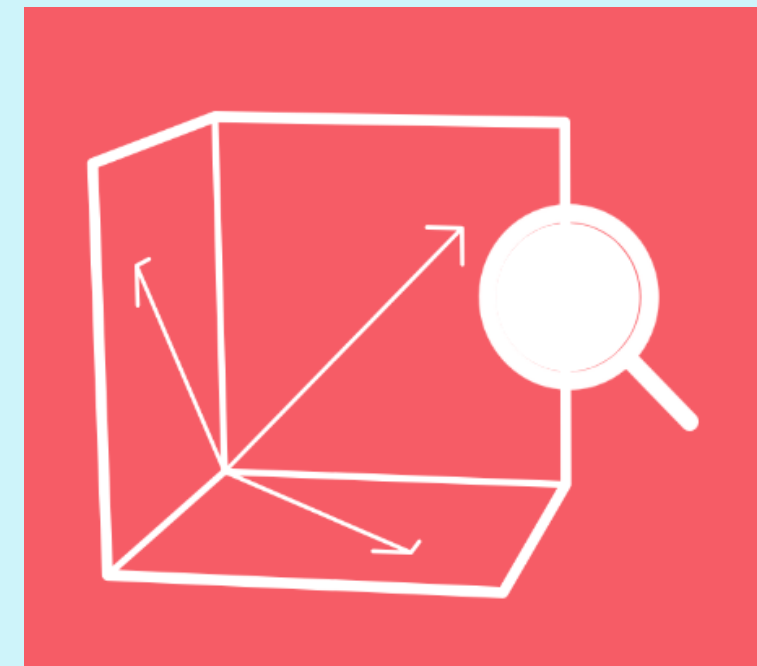
Searching Approach

Keyword Search



Look for **exact words**.

Semantic Search



Look for **similar meaning**.

Searching Approach

Keyword Search or Sparse Vector

- Bag of words
- TF-IDF
- BM25

Semantic Search or Vector Search

- Use embedding model then store in DB.

Searching Approach

Keyword Search

- Very precise for **exact terms**, IDs, names, codes
- **Easy** to explain why a result matched

Semantic Search

- Understands **meaning**, not just exact words.
- Handles **synonyms** and paraphrases well

Precision and Recall

Precision

Measures how many of the returned documents are relevant

Relevant Retrieved / Total Retrieved

Recall

Measures how many of the relevant documents are returned

Relevant Retrieved / Total Relevant

Precision and Recall

- Precision

- ที่ดึงมาได้ทั้งหมด, มีที่อื่นที่เกี่ยวข้อง
- ลงโทษ → **ดึงตัวไม่เกี่ยว**

- Recall

- จากทั้งหมดที่เกี่ยวข้อง, ดึงมาได้กี่อัน
- ลงโทษ → **การ leave out**

RAG IMPROVEMENT

Chunking

Why chunk ? Insert full document เเลยได้ม๊ยะละ



Token Limits



**Improved
Relevancy**



**LLM only sent
relevant context**

Chunking Strategy

- Fixed Size Chunking
 - Overlapping Chunking
 - Recursive Character Splitting
 - Splitting on Different Characters
- Semantic Chunking
 - Language Based Chunking
 - Context-Aware Chunking
- Parent-Child Chunking

Chunking Strategy

- **ระวัง ภาษาไทย**
 - *Problem:* การตัด กลางคำ
 - *Solution:*
 - Tokenize → แยกด้วย pipe |
 - pythainlp: attacut

Query Parsing

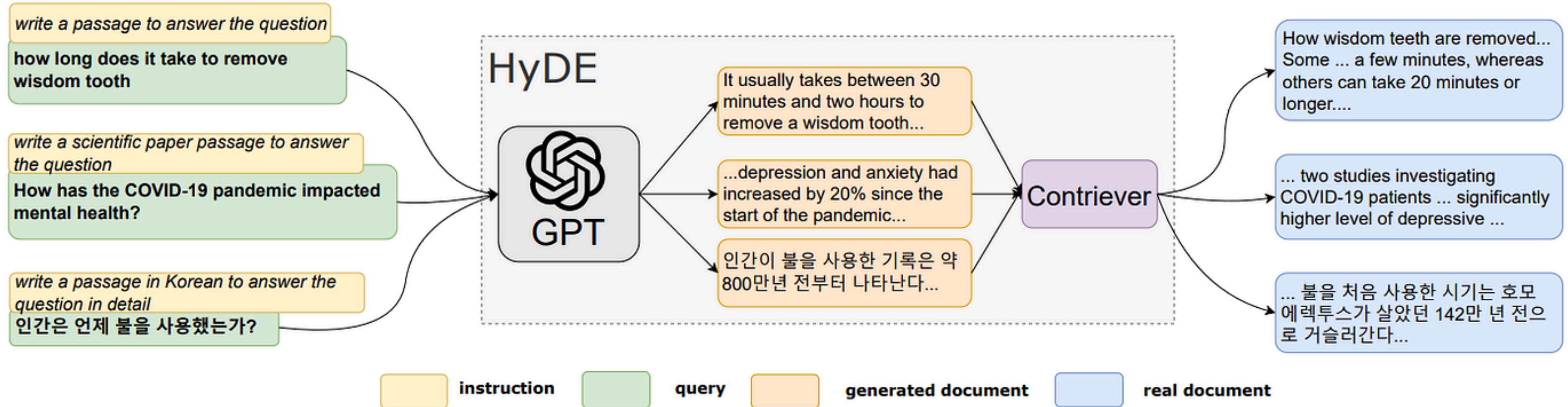
- Query Rewriting
- Named Entity Recognition (NER)
- Hypothetical Document Embeddings (HyDE)

Query Parsing

ORGANISATION LOCATION DATE PERSON WEAPON

The **ISIS**_{ORG} has claimed responsibility for a suicide bomb blast in the **Tunisian**_{LOC} capital **earlier this week**_{DATE}, the **militant group**_{ORG}'s **Amaq news agency**_{ORG} said on **Thursday**_{DATE}. A **militant**_{PER} wearing an **explosives belt**_{WEAPON} blew himself up in **Tunis**_{LOC}

Query Parsing



ไม่สอน แต่อยากให้ได้รู้ไว้

- Bi-encoders
- Cross-encoders
- ColBERT (Contextualized Late Interaction Over BERT)

RAG Improvement

- Improve recall
 - Query Rewriting
- Improve Precision
 - Reranking

RAGAS Library



ragas score

generation

faithfulness

how factually accurate is
the generated answer

answer relevancy

how relevant is the generated
answer to the question

retrieval

context precision

the signal to noise ratio of retrieved
context

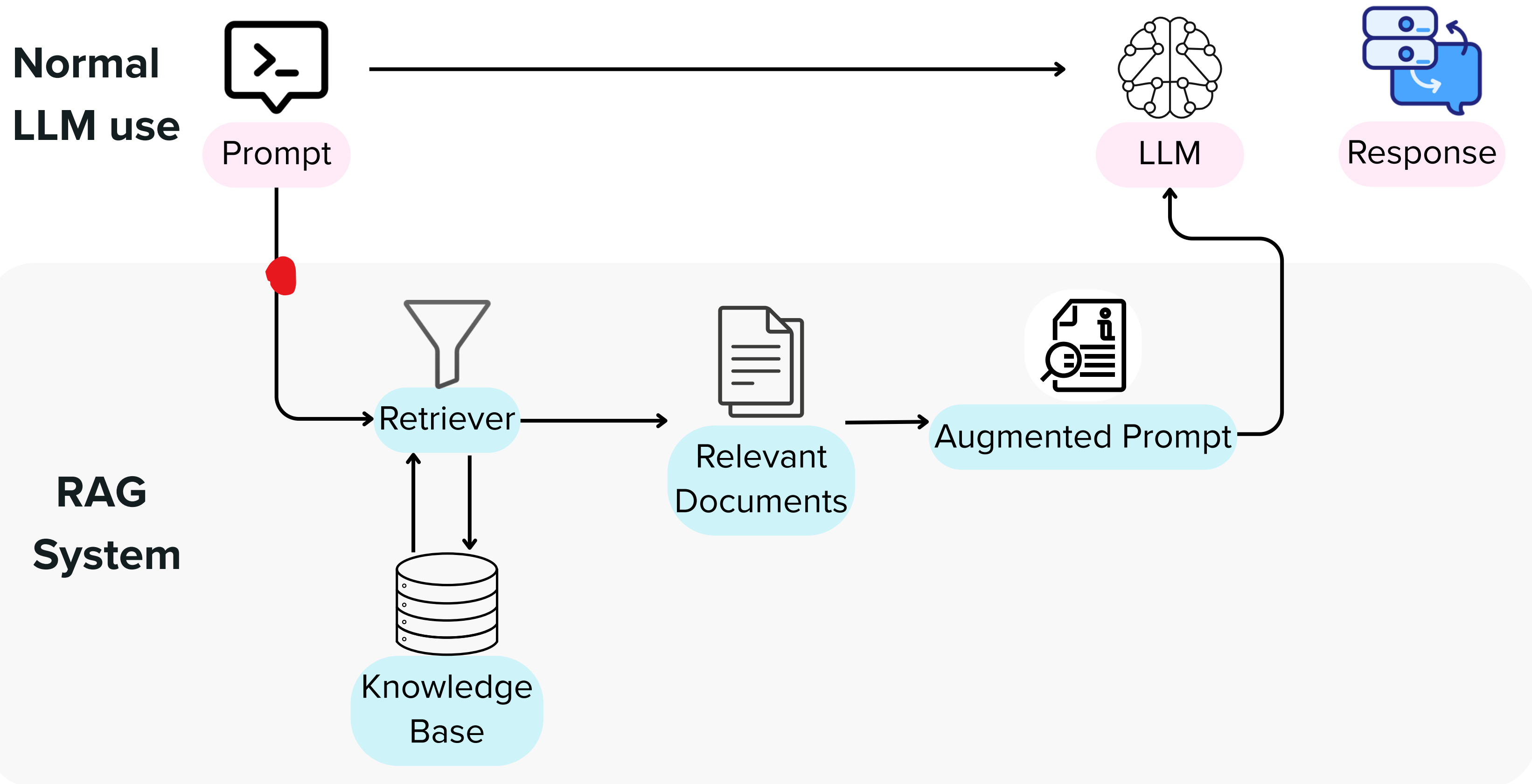
context recall

can it retrieve all the relevant information
required to answer the question

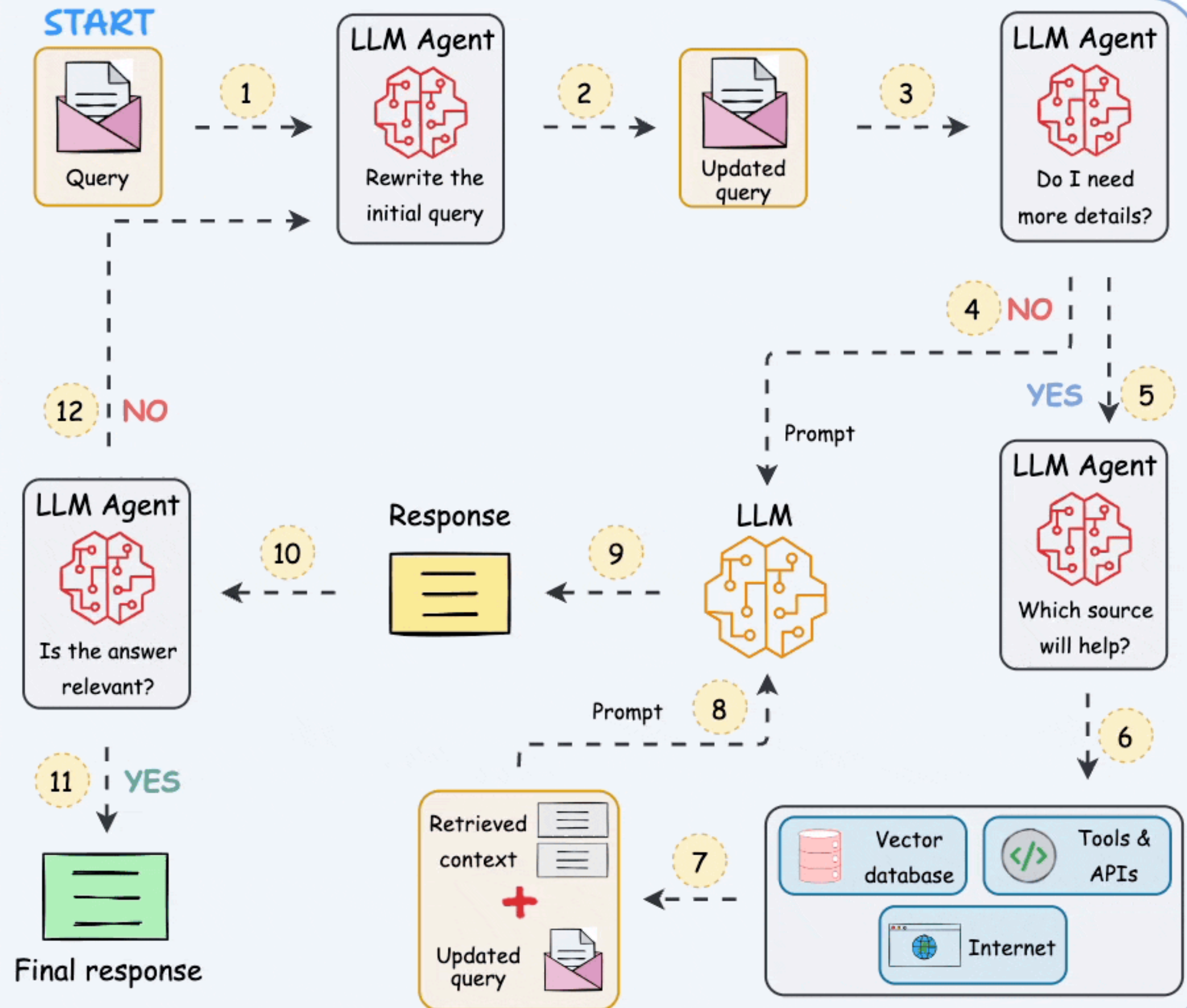
แถม ลำดับ

- **LangGraph Basic Agent Coding**
- Tracing: Phoenix
- Quantization
- Latency
- Security
- Multimodal RAG

AGENTIC RAG



Agentic RAG



Q&A

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